Firing rates,

turning curves, & receptive fields

Computational Neuroscience

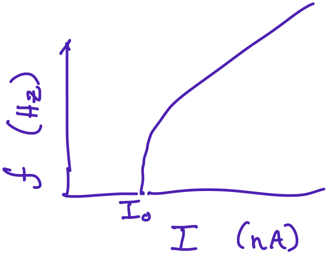
University of Bristol

mrule

Learning outcomes:

- What are firing rates?
- How do we measure them?
- What is rate coding; how do we model it?
- Tuning curves and receptive fields

LIF NEURON f-I CURVE

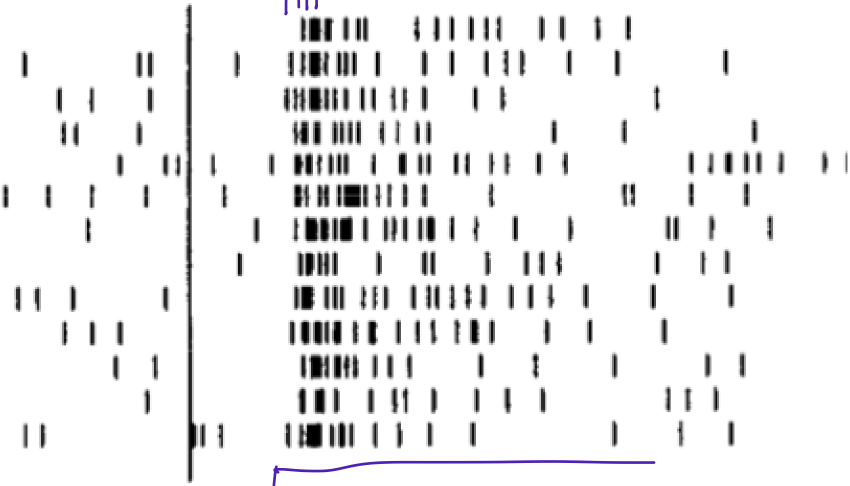


Firing rates

BE PREPARED TO SKETCH SOMETHING
LIKE THIS ON EXAM

What are firing rates?

Δt



Motter, J Neurophysiol, 1993

todo: get horizontal scale?

Raster plot of spikes from a single monkey visual cortex neuron from repeated presentations (each row)

Properties of neural firing rates "r"

Non-negative $r \geq 0$ and continuous $r \in \mathbb{R}_{\geq 0}$

*Play with the Hodgkin Huxley lab to see how you'd need to change voltage-gated conductances to support this.

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Usually latent variables rather than directly measurable

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- Modelling abstraction simplifying spiking to a probability/frequency/rate

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Varies systematically across brain regions

- Typically low on average (0.1–10 spikes per second; Hz)
- Pyramidal cells in superficial somatosensory cortex fire ~0.1 Hz
- Purkinje neurons in the cerebellum fire ~50 Hz spontaneously
- Some cells can sustain rates up to >~1000 Hz*

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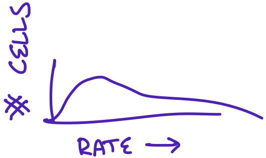

Mean rates heterogeneous, even within a population of neurons of the same type

- Typically ≈ log-normal distribution

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How do we measure firing rates?

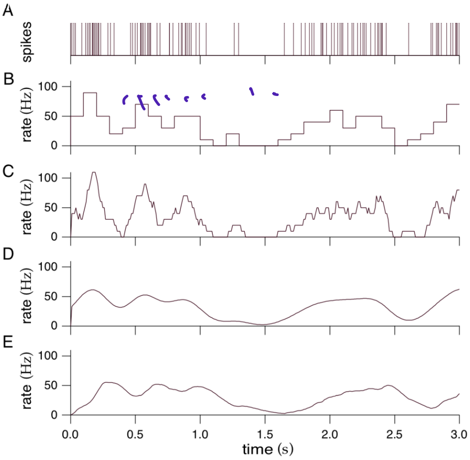


Figure 1.4: Firing rates approximated by different procedures. A) A spike train from a neuron in the inferior temporal cortex of a monkey recorded while that animal watched a video on a monitor under free viewing conditions. B) Discrete-time firing rate obtained by binning time and counting spikes with $\Delta t = 100$ ms. C) Approximate firing rate determined by sliding a rectangular window function along the spike train with $\Delta t = 100$ ms. D) Approximate firing rate computed using a Gaussian window function with $\sigma_t = 100$ ms. E) Approximate firing rate for an α function window with $1/\alpha = 100$ ms. (Data from Baddeley et al., 1997.)

DOESNT SHOW ON IPAD

How do we measure firing rates?

\# spikes / # neurons in the population

- Out of 1000 excitatory cells, what fraction spiked in the last $\Delta t = 1$ ms?
- "Population rate"
- Defined for low temporal latency, blurs out roles of single cells

\# spikes / time (time-average mean rate)

- How many spikes did this neuron fire in the last $\Delta t = 100$ ms?
- Defined for single neurons, over sufficient time (or trial replicas)

Some mixture of the above?

Draw a picture for each on the right

Other quantities related to rates

Spike ***counts*** $k \in \mathbb{Z}_{\geq 0}$

- \# spikes you actually saw in a population and/or over a time interval

Firing ***intensity*** $\lambda(t) \in \mathbb{R}_{\geq 0}$

- A firing rate that varies continuously in time and is defined for $\Delta t \to 0$
- Mathematical abstraction to represent probability of spiking

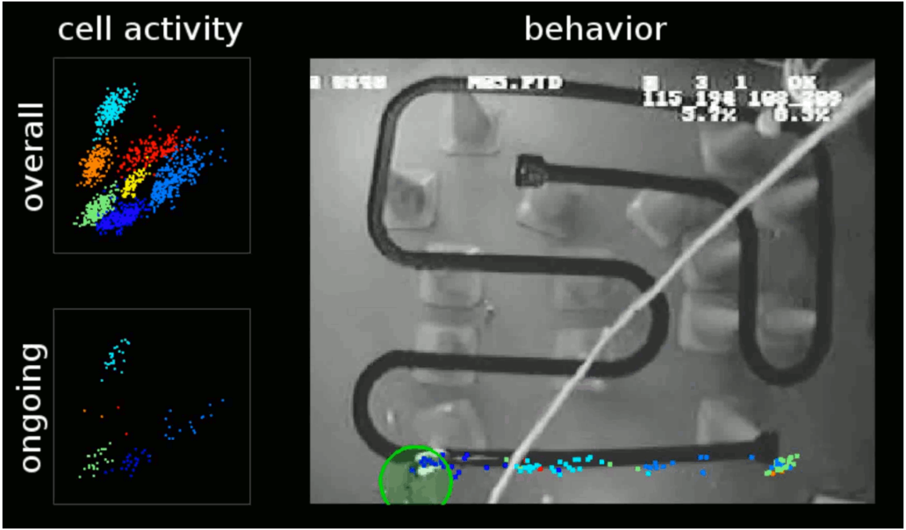
$$\Pr\left(\text{spike count } k = \int_t^{t+\Delta t} y(t)\ \mathrm{d}t\right) \sim \text{Poisson}\left(\lambda_{\Delta t}(t) = \int_t^{t+\Delta t} \lambda(t)\ \mathrm{d}t \approx \lambda(t) \cdot \Delta t\right)$$

- Point-process modelling: Week 7, extended coursework

Rate coding in single neurons

Rate coding

This Video Is Now Downloaded To
The Week 5 GitHub Slides Folder



Hippocampal place cells recorded in the Wilson lab at MIT

Rate coding

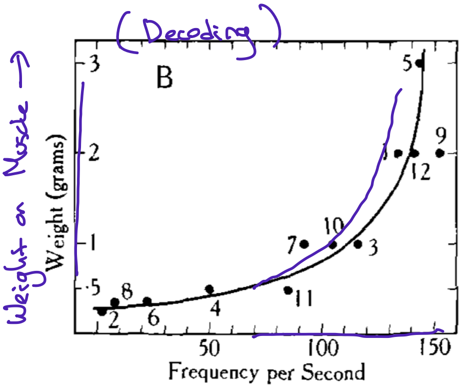
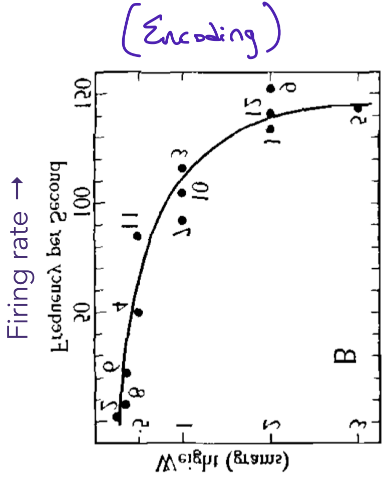


FIG. 25. FROG'S STERNO-CUTANEOUS PREPARATION. RELATION BETWEEN INTENSITY OF STIMULUS (WEIGHT ON MUSCLE) AND FREQUENCY OF DISCHARGE.



Weight on muscle →

The Basis of Sensation, Adrian, 1928, via L. Aitchison's slides.

When a neuron's response to a stimulus or applied current is reproducible, we can repeat the same experiment many times (many "trials"), to define a rate response. Instead of averaging over $\Delta t = 100$ ms, we could average over 100 trials at $\Delta t = 1$ ms, to get a finer rate resolution.

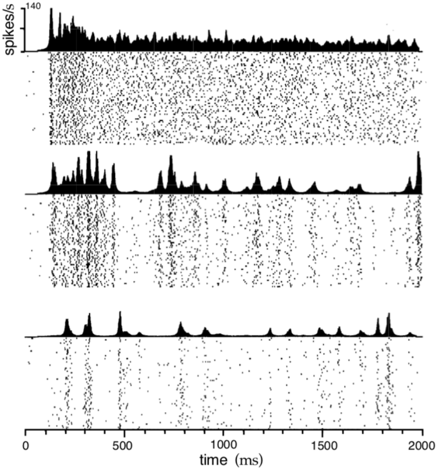


Figure 1.19: Time-dependent firing rates for different stimulus parameters. The rasters show multiple trials during which an MT neuron responded to the same moving random dot stimulus. Firing rates, shown above the raster plots, were constructed from the multiple trials by counting spikes within discrete time bins and averaging over trials. The three different results are from the same neuron but using different stimuli. The stimuli were always patterns of moving random dots but the coherence of the motion was varied (see chapter 3 for more information about this stimulus). (Adapted from Bair and Koch, 1996.)

Rate coding over time

From Dayan & Abbott, 2001

Reality check: trials are rarely strictly reproducible *in vivo*, and statistical methods to dissociate per-trial from mean-rate effects are an area of active research.

Rate coding: Remember

Sometimes rate coding makes sense:

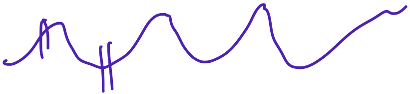
- Spikes from α motor neurons are low-pass filtered by the dynamics of the musculoskeletal plant
- Average rates of many similarly-tuned neurons to variations in brightness in the visual system
- Intensity of pressure on the skin, averaged over nearby mechanoreceptors
- Area MT response to motion (trial averaged)

plant = thing being controlled, in control theory

Rate coding: Remember

Sometimes rate coding makes no sense:

- Sometimes exact spike timing can be very important
- **Spike timing code**
  - Interaural time difference for sound localization
  - Nucleus laminaris (birds) / medial superior olive (humans)
  - Spike arrival time difference $\mathcal{O}(10\ \mu s)$
- **Phase code**
  - Hippocampal neurons' spike phase relative to "theta" oscillation encodes time



Rate coding: Remember

Some examples are subtle

- A sensory stimulus can trigger a rapid, feed-forward wave of neural activity, and be recognized with only time enough for neurons to fire a single spike. Here, a *temporal* rate does not make sense — but a *population* rate code might.

Tuning curves, receptive fields, ~~response fields~~

Tuning curve vs. receptive field

Some neurons in the brain respond to certain aspects of sensory stimuli by firing action potentials. Tuning curves and receptive fields are ***compact descriptions*** of these stimulus-response relationships.

On an exam, a "tuning curve vs. receptive field" question would usually be restricted to e.g. vision, where the distinction is most clear/meaningful.

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- A description of its firing rate as a function of some property of the stimulus
- E.g. "This neuron fires most to yellow–blue edges oriented at 45°"

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- Subset of the stimulus space that the neuron responds to
- *Primarily* used for sensory inputs that have a natural spatial organisation
  - Region of the visual field a retinal ganglion cell responds to encodes
  - Region of skin a mechanosensor covers

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"Receptive/ves field" is roughly analogous to a probability distribution's support (inputs $x \in \mathcal{X}$ where it makes sense to define a value), whereas a "tuning curve" would be the specific values $\Pr(x)$ of this distribution at the supported inputs.

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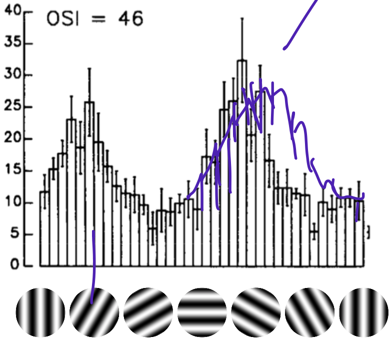
Tuning curves and receptive fields



Chapman & Stryker, J Neurosci, 1993

Field & Chichilnisky, Annu Rev Neurosci, 2007

Overlaid on Clifton Bridge

The bar chart is a **tuning curve**. The neuron is tuned to a ***preferred orientation***.

The small part of the visual world that this neuron "sees" is its **receptive field**.

Modelling rate coding in single neurons

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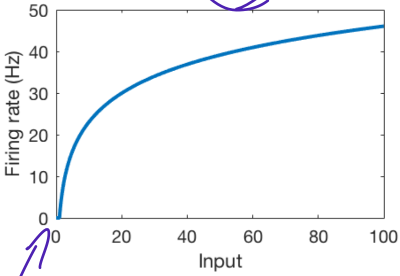
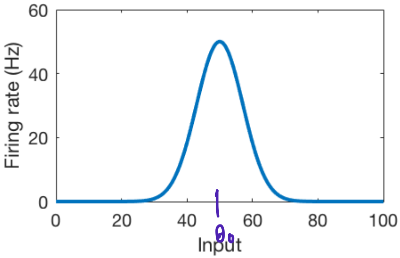
$$r(x) = \alpha \log(x)$$



e.g. mechanoreceptors, visual contrast

Floating Point Neuron

$$r(x) = \alpha e^{-\left(\frac{x-\mu}{\sigma}\right)^2}$$



e.g. edge orientation, place fields

Weber's law

Animals are sensitive to relative $(\Delta x / x)$, not absolute (Δx) changes.

We will revisit rate-based statistical models of stimulus encoding and decoding in week 7 in relation to the extended coursework.

Modelling rate coding in neural populations

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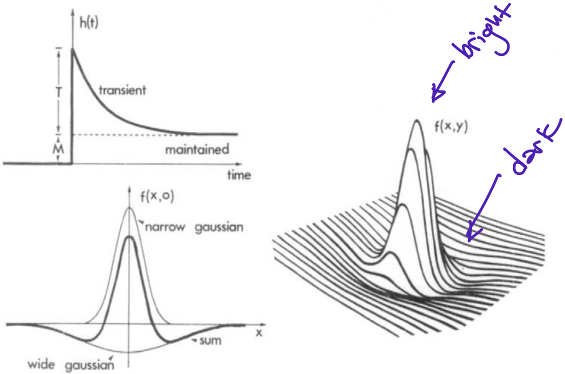
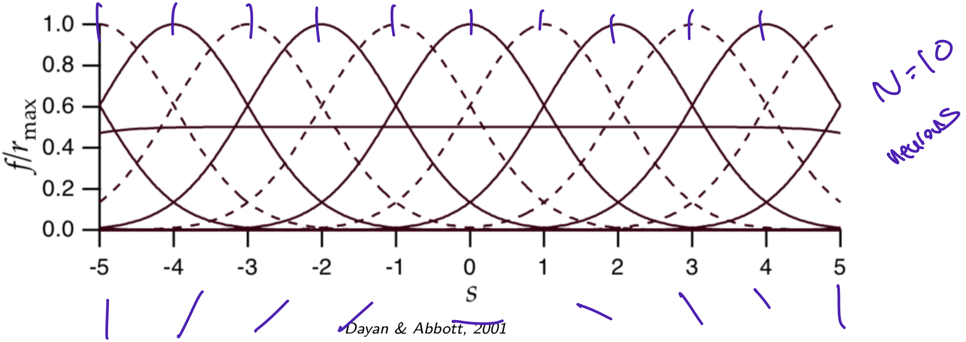


Rodieck, Vision Res, 1965

$$f(x,y) = \frac{g_1}{\pi\sigma_1^2}\exp\left(-\frac{x^2+y^2}{\sigma_1^2}\right) - \frac{g_2}{\pi\sigma_2^2}\exp\left(-\frac{x^2+y^2}{\sigma_2^2}\right)$$

Difference of Gaussians model captures the 'ON' centre and 'OFF' surround ***typical of retinal ganglion cells***.

Modelling rate coding in neural populations



Dayan & Abbott, 2001

A stimulus 's' can be represented by the collective activity of a population of neurons, each with a different preferred stimulus (i.e., their tuning curves are centred at different locations).

In some cases (e.g. when a stimulus space varies smoothly and continuously is well-covered by a neural population), it can make sense to pretend that we have an infinite population of neurons with a cell for each possible S. This is a *neural field model* and we will cover this in more detail later.

Firing Rates: Summary

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A RATE is a continuous, non-negative quantity that abstracts the number of spikes per unit time.

Calculation of inter-aural time differences in the brainstem uses a SPIKE TIMING CODE, and is POORLY modelled as a rate code.

Muscle force takes a low-pass average of α-motorneuron spiking and is WELL modelled as a rate code.

Place cells' spiking relative to the theta oscillation is an example of a PHASE CODE

A (e.g. visual) neuron's

- RECEPTIVE FIELD is the sub-region of a sensory space is responds to.
- TUNING CURVE is the relationship between spiking and stimulus features in said sub-region.

Weber's law says that animals are typically sensitive to RELATIVE variations in stimulus intensity

(short answer): Briefly describe what we mean by "population rate code":

Populations of neurons with different tuning curves work together to encode and represent the possible values of a sensory stimulus.

Firing Rates: Summary

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end

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